

Hybrid CNN-GNN Architecture for Infrastructure Classification in Satellite Imagery

by
MALEESHA LIONEL
20960793

Supervised by
MS CHATHURA SOORIYA-ARACHCHI

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DECLARATION

I, Lelwala Guruge Maleesha Diluminda Lionel declare that I am the sole author of this Project; that all references cited have been consulted; that I have conducted all work of which this is a record, and that the finished work lies within the prescribed word limits.

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ABSTRACT

List of Abbreviations

CNN - Convolutional Neural Networks
GAT - Graph Attention Network
GCN - Graph Neural Network
GNFA - Graph Neural Aggregation
GNN - Graph Neural Networks
HSI - Hyperspectral Image
IP - Indian Pines Dataset
KSC – Kennedy Space Centre Dataset
PCA - Principal component analysis
RGB - Red, Green, Blue
RM - Research Method
RO - Research Objective
RQ - Research Question
RS - Remote Sensing
SLIC - Simple linear iterative clustering

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1. INTRODUCTION

In the era of artificial intelligence, the data science community is becoming increasingly interested in remote sensing, and particularly satellite imagery, as they work to enable machines to recognise their surroundings through the classification of satellite images. For both military and civilian uses, imaging satellites gather, examine, and process photographs of the surface. The subjects of biodiversity, meteorology, fisheries, agriculture, cartography, geology, land use planning, oceanography, and military, among others, can all benefit from the usage of satellite imagery. Instead of just having an image of a location, satellite image classification seeks to convert satellite images into useful information.

Remote sensing images help manage and understand cities' growth by giving precise and detailed images (Nita Nimbarte, 2024). CNNs automatically extract and recognise features, eliminating the need for manual feature extraction and selection techniques. (Hosny et al., 2021, p2). Although CNNs are successful in terms of learning local feature visual patterns, they fail in capturing the larger spatial context of an image, i.e., how distant, although related, objects are placed in relation to one another. This weakness is particularly crucial in RS, where there can be irregular and discontinuous distributions of objects such as roads, buildings, and industrial regions. CNNs can only focus on small, fixed parts of an image at a time, which makes it hard for them to understand complex spatial patterns (Li et al, 2020). In recent years have discovered that GNN-based CNN models outperform CNN-only models because they can deal with irregular regions on images and capture both global and structural dependencies (Wang et al, 2024, p2).

GNN networks are good at working with data that is not arranged in a regular grid format since GNNs treat image parts or areas as nodes and apply graph-based convolutions to model their relationships. This is especially good for RS images, as the structures of roads, buildings, and zones there are very complex for CNNs to represent (Zhang et al, 2024). In addition, GNNs can work well even when some labels are missing and help improve model performance when there is only a small amount of labelled data (Hu et al, 2021). For this reason, combining CNNs and GNNs can be a great option since CNNs are strong at finding features and GNNs handle learning relationships well (Thangamariappan et al, 2024).

The work introduces a model that uses both CNNs and GNNs to classify the types of infrastructure, Residential, Industrial, and Highway, that are present in the EuroSAT dataset. It uses spatial connections to help make better classifications, dealing with issues such as imbalance in the training data, differences at the pixel level, and the shortcomings of simple grid structures in classifying images. The project intends to build a scalable and understandable method for classifying infrastructures using CNNs and GNNs in remote sensing.

2. BACKGROUND

Satellite image remote sensing is becoming a valuable means of land use and infrastructure development monitoring. The development of satellites allows increased spatial resolution and imaging frequency, which makes it possible to monitor urban development, networks of roads,

and transformations in land cover over time. There are a number of published datasets providing remote sensing research, such as EuroSAT (Helber et al., 2018), LEVIR-CD (Kömürcü and Petkevičius, 2024), and others, e.g., Google Earth Engine (Shelestov et al., 2017). EuroSAT provides labelled multi-spectral satellite images of the Sentinel 2 Satellite with a wide range of land cover classes of images, such as Residential, Industrial, and Highways, which are relevant as per the objectives of this study of infrastructure classification. LEVIR-CD is primarily aimed at indicating change in urban environments. In contrast, Google Earth Engine is a cloud-hosted tool for geospatial data processing, enabling change detection. Yet, it would not serve to acquire a labelled dataset to conduct change detection. Thus, EuroSAT was chosen due to its easy accessibility, well-labelled data, and smaller data size, as well as its direct applicability to the classes of infrastructure.



Figure 1: Levir-CD and EuroSat images from the Levir-CD and the EuroSat Dataset

The images taken by satellites are not the same as photos taken by humans. Unlike regular photographs, the layouts of roads and buildings can be very complex, which makes it hard to represent them with standard convolutional grids.

Graph Neural Networks (GNNs) have proven to be an effective method of modelling non-regular grid format, spatial relationships in remote sensing data to address these shortcomings (Hong et al., 2021). GNNs offer better advantages when compared to CNNs in learning structural and relational representations. While CNNs are better at visual tasks, and require numerous layers and high computational power (Jiao, Zhang and Li, 2024). On the other hand, GNNs can efficiently explore topological relationships in graph data with fewer layers, which means less computational power (Jiao, Zhang and Li, 2024).

A hybrid architecture that combines CNNs and GNNs can take advantage of their complementary strengths. CNNs are good at extracting rich features, and GNNs are good at spatial reasoning. This combination enhances classification outcome in terms of infrastructure, including Residential, Industrial, and Highway areas, as it deals with pixel-level variability, class imbalance, and complicated spatial patterns. The proposed hybrid model in the project would offer a scalable and interpretable infrastructure classification solution with practical implications to urban planners, transportation authorities, and environmental agencies.

3. PROBLEM STATEMENT

Reliable classification of satellite images is essential for disaster management, environmental monitoring, and urban planning (Mukhedkar et al., 2023, p1). However, traditional Convolutional Neural Networks (CNNs) require high computational demand, a large number of parameters, and a long training time for classification tasks (Hasan and Islam, 2025, p1). This limitation becomes significant in remote sensing (RS) images, where objects such as roads and buildings are often scattered, irregularly shaped, or viewed from varying angles.

Recent research explores combining CNNs and GNNs to enhance image analysis tasks. CNNs excel at extracting local features but struggle with complex spatial relationships, while GNNs can model intricate connections in structured data (Thangamariappan et al., 2024).

A hybrid GNN-CNN architecture can leverage the strengths of both networks to extract features and capture the spatial relationships in satellite images, which could make it drastically accelerate the accuracy and robustness of land use classification, especially in situations of inconsistent structures and small training samples.

3.1 Aim

The aim is to perform infrastructure classification in satellite imagery by developing and evaluating a hybrid CNN-GNN model that uses both local visual features and spatial relational context, using the satellite image data. The model will also be easier to apply in real-life remote sensing, since it works well under fewer labelled datasets with fewer computational needs.

3.2 Research Question

How does integrating a GNN component into a CNN architecture influence classification accuracy and spatial consistency for infrastructure detection in EuroSAT satellite imagery ?

3.3 Objectives

1. Design and implement a hybrid CNN-GNN model that integrates feature extraction from CNNs with spatial relationship modelling using GNNs, tailored to the EuroSAT dataset.
2. Train the hybrid model on labelled satellite images for infrastructure classification into Residential, Industrial, and Highway categories, incorporating preprocessing strategies to address class imbalance and spatial variability.
3. Compare the performance of the hybrid CNN-GNN model with baseline CNN-only models using metrics such as accuracy, F1-score, AUC-ROC, and precision-recall under equal training resources.
4. Critically assess the scalability, interpretability, and limitations of the proposed hybrid architecture in the context of infrastructure classification for real-world satellite applications.

4. LITERATURE REVIEW

This review of the literature provides an overview of recent literature on the classification of infrastructure using satellite imagery, the advantages and disadvantages of CNNs and GNNs, and how a combination of the two models can improve the performance on more challenging remote sensing tasks.

4.1 Infrastructure Classification in Satellite Imagery

Satellite imagery-based infrastructure classification has not only been among the essential applications of remote sensing, but it also plays essential functions in urban and disaster planning, as well as environmental monitoring. This is a complex task due to the multiscale characteristics of infrastructure components, and they can be anything from infrastructures noted per building to huge urban networks and transportation corridors.

The traditional methods of infrastructure classification were based on interpreting methods and rule-based algorithms that were mainly manual and could not cope with the growing complexity and size of modern satellite data. High-resolution satellite image classification using conventional techniques has been limited due to complex image characteristics like spectral signatures, texture, shape, and spatial relationships. (Adegun, Viriri and Tapamo, 2023, p21).

The analysis of satellite imagery in infrastructure has a different challenge brought about by different scales, orientations, and geometrical relationships (Xia et al., 2010). Convolutional neural networks have demonstrated effectiveness in finding critical infrastructure with a high rate of accuracy, exceeding 90% under the six infrastructure facility types (Elliott et al., 2022, p1). Nevertheless, spectral similarities among infrastructure types may cause poorer accuracy levels in some categories (Elliott et al., 2022, p10).

4.2 Convolutional Neural Networks

The introduction of Convolutional Neural Networks has transformed the domain of computer vision and has established itself as the default tool of any image processing task in diverse fields. The structure of CNNs is modelled after the system of the visual cortex in animals and is specifically developed to work with grid-like data, e.g., images. CNNs have more than one layer, which are composed of convolutional, pooling, and fully connected layers, with different purposes on each layer in the feature learning process.

The main idea of CNNs is that their feature hierarchy in different space dimensions can be obtained automatically as a result of backpropagation. Contrary to conventional machine learning methods, where feature engineering must be done by hand, CNNs are able to learn useful features themselves, working directly on raw pixels. This ability makes them especially appealing to difficult visual recognition problems in which the best features are not necessarily obvious or easy to hand-design.

The convolutional layer, which gives CNNs their name, convolutional neural nets, is an operation that uses small filters (kernels) to detect local identifications like edges, corners, and textures by sliding them across the input picture. Such filters are acquired in the course of training, and they get specialised in the detection of special patterns that pertain to the task. The structure of layers in CNNs enables the basic or simplistic features, such as edges and gradients, to be detected in the lower levels of the hierarchy, and greater complex patterns and objects are identified by combining the simple features in the deeper levels.

Pooling layers in CNNs have several functions: to decrease the spatial dimensions of feature maps, to ensure translation invariance, as well as to reduce overfitting of the model by lowering network parameters. The most popular types of pooling are max pooling and average pooling, which summarise local neighbourhoods of feature maps. This dimension reduction is essential in terms of control over the complexities of computation and retention of the most important information.

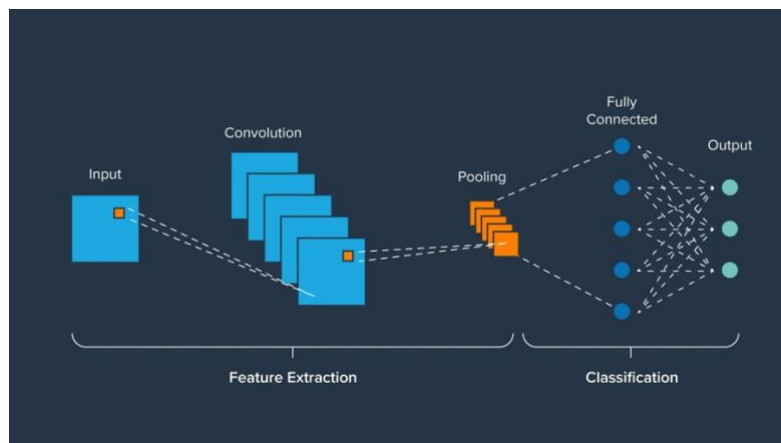


Figure 2: CNN Architecture - (Exxact Corp. Blog | HPC, AI, & Data Center Insights)

4.3 Limitations of CNNs in Remote Sensing

The latest studies emphasise the fact that convolutional neural networks (CNNs) fail at long-range spatial dependency representation when it comes to examining a satellite image. To resolve this, a few methods have been brought forward. RoadTagger integrates CNNs with Graph Neural Networks to incur spatial correlations on road networks to enhance the accuracy of attribute inference.(He et al., 2020, p2). The localised receptive fields in CNNs limit their ability to comprehend the global context and spatial relationships beyond the pixel neighbourhood since they process the images in a localised working memory.

The use of CNNs becomes problematic within the irregular spatial domain due to the design and convolution of a kernel filter being non-trivial (Edwards and Xie, 2016, p1). Another significant limitation is that the traditional convolutional neural network (CNN), characterised by fixed-sized windows, needs help capturing intricate spatial and temporal relations at the pixel level, and the immutability of CNN filter weights post-training further restricts input fluctuations(Habib, Qureshi and ishfaq, 2024, p1). High information content of satellite images alongside high computational calculations needed by CNNs, which makes performance issues very crucial (Laban et al., 2018, p1).

In this study, these shortcomings are overcome by incorporating a Graph Neural Network (GNN) into the CNN architecture. Whereas the CNN aims at extracting the local picture features (textures, shapes, and edges), the GNNs model patterns of different areas in the picture with boundaries between them, making graph nodes interconnected according to the adjacency or similarity. This allows the proposed hybrid CNN-GNN architecture to learn both local feature representations and longer choices of spatial relationships better than a single CNN architecture for better scene-level classification accuracies and higher spatial consistencies in complex infrastructure scenes.

4.4 CNNs for Satellite Image Classification

CNNs have been effective for satellite image analysis since they have become efficient tools for image processing tasks. By utilising several layers that automatically learn to identify features like edges, shapes, and textures, CNNs are specifically made to recognise patterns in images. Because of this, CNNs are ideally suited for tasks involving the analysis of satellite images, where it is crucial to distinguish between different types of land cover, track changes over time, and segment regions of interest (Electronics and Communication Ahalia School of Engineering and Technology, Palakkad, Kerala, India et al., 2024, p1). The main strength of CNNs lies in their ability to learn hierarchical feature representations automatically, which those encapsulated in raw pixel data and thus, they are especially adaptable when dealing with complex visual patterns like the ones found in satellite imagery.

CNNs' architecture offers a variety of benefits for processing satellite images. Rapid, economical, and accurate analysis and feature extraction from remote sensing data can be achieved by using convolutional neural networks (CNNs), which are capable of directly extracting features from vast volumes of image data and excel at using the semantic features of image data. (Song et al., 2019, p1). Also, CNNs can handle complicated image data, extract multi-scale and multi-level features, and produce high-precision classification results because of their unique convolutional layer, pooling layer, and fully connected layer architecture. (Gu, 2025, p1)

The latest trends in CNN-based methods applied to satellite imagery are related to multi-scale feature extraction, attention mechanisms, and transfer learning approaches. Such innovations have led to considerable enhancements in the accuracy of classification, especially in complex scenes having a multiplicity of infrastructure of various types and environmental conditions.

4.5 Graph Neural Networks

Graph Neural Networks (GNNs) are deep learning frameworks that work with data structured as graphs, which means they can be especially useful in extracting the spatial dependencies that are prevalent in remote sensing spatial data. GNNs differ from CNNs in that, whereas CNNs use grid-based structures, the irregular patterns occurring in space can be modelled by viewing image regions, e.g., superpixels or segmented patches, as nodes with edges corresponding to their spatial locations and representing spatial proximity or feature similarity.

GNNs can also learn about both the local and global dependencies as each node accumulates contextual knowledge through iterative message-passing operations to aggregate information about its neighbours. Such relational learning ability can be particularly useful towards classifying infrastructure where roads, buildings, and zones combine to create networks of

interconnected structures that cannot be simply defined by conventional pixel-based approaches. This difference in visual feature extraction ability between GNNs and CNNs should not be significant since the GNN has lower fine-grained visual feature extraction ability; however, the GNN has a better sense of modelling spatial context, making it part of such a combination that will be very suitable in a hybrid model to analyse complex remote sensing data.

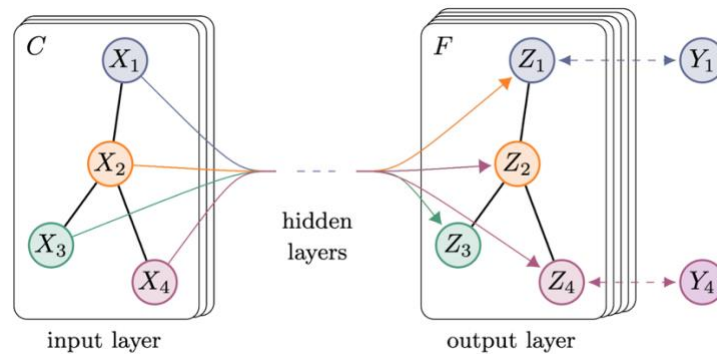


Figure 3: Figure - Graph Neural Network (Kipf and Welling, 2017)

4.6 Graph Neural Networks (GNNs) in Remote Sensing

GNN has emerged as a common technique for semi-supervised classification, and it has been incredibly popular when applied to hyperspectral images. (Niruban and Deepa, 2023, p1). GNNs are a significant innovation, which makes DL enter the non-Euclidean world. According to (Zhao et al., 2025, p1).

GNNs work by creating graphs where each node is connected to other nodes via edges, and then aggregating information across these connections. This allows GNNs to capture long-range contextual information and improve the smoothness of features across neighbouring spectral regions, which helps improve classification accuracy (Wang et al., 2021, p1; Zhao et al., 2025, p2).

A graph filter or single graph neural network has been typically used in the earlier GNN-based approaches, and it is used to obtain HSI characteristics (Niruban and Deepa, 2023, p1). With the possibility to model both spatial and temporal relationships in terms of graph structures, new opportunities have emerged in the dynamic monitoring and analysis of infrastructure.

With the growing number of GNN-based approaches, the possibility to combine spatial, spectral, and temporal data makes their use as a remote sensing tool immensely effective. They also allow better classification with complex and irregular scenes since they have a flexible graph structure, and there is the possibility of real-time monitoring and change detection. That is why GNNs are very applicable to remote sensing tasks, where it is crucial not only to learn the local features but also to remember the global connections.

4.7 Comparative analysis of CNN, GNN, and Hybrid Approaches

The discussion of CNN, GNN, and hybrid methods allows understanding the specific strengths and weaknesses of the analysis of satellite imagery. Although convolutional neural networks have been successful in their applications to the recognition of objects in pictures and videos,

the current GNN architecture presents a brand new way of solving diverse image and video comprehension tasks that might involve spatial relations awareness.

CNNs are better in local feature extraction and pattern recognition, and the fact that CNNs can automatically learn hierarchical representations on raw source data makes them great at identifying individual infrastructure components.(Wang et al., 2021, p1,2). However, their low capability in modelling the global spatial relations comes to light when one is concerned with the complex infrastructure networks, where the connectivity and spatial context must be understood.

The analysis above shows that hybrid solutions tend to excel compared to the current standalone architectures because they help overcome the weakness of the corresponding architectures and introduce feature complementarity.

4.8 Hybrid CNN-GNN Architectures

Hybrid CNN-GNN networks are the next state-of-the-art approach in deep learning works with satellite imagery, combining the best of both worlds. Such architectures follow a convention of using CNNs to pre-extract features from raw satellite imagery and GNNs to represent the spatial relationships, thereby enabling classification that relies not only on visual features but also on spatial context.

The integration approaches for hybrid models are dependent on the type of requirement in the application. Sequential integration entails introducing some of the extracted features by CNN as node properties utilizing a graph system and followed by a GNN classification. Parallel integration performs the same input in both CNN and GNN branches, whereas a combination is performed at the decision level.

GNNs-CNN hybrid approaches benefit in terms of performance due to their synthesis of the advantages of both GNNs and CNNs (Thangamariappan et al., 2024). This ability to learn representation comprehensively adapts them very well to the tasks of complex infrastructure classification, where it is necessary to capture both the properties of individual elements and the structure of connections between elements on a network-level basis.

Future studies of hybrid CNN-GNN designs would involve working on the problem of computational efficiency, introducing newer, more sophisticated fusion schemes, and breeding ad-hoc architectures that can accommodate specific remote sensing tasks. Explainable AI has also impacted the design process of hybrid models, where researchers have encompassed attention mechanisms and interpretability capabilities in them.

4.9 Existing Work

Table 1 summarises recent research in using CNNs, GNNs, and hybrid networks in satellite image classification. Current methods demonstrate the importance of CNNs in local feature representations, as well as the influence of GNNs in capturing spatial relationships.

Paper Title	Architecture	Datasets Used	Contribution	Limitation/Note
A Deep Neural Network Combined CNN and GCN for Remote Sensing Scene Classification(Liang, Deng and Zeng, 2020)	CNN + GCN	AID, NWPU-RESISC45, DOTA	Combines CNN feature maps with graph-based spatial reasoning	Scene-level classification only
Kolmogorov-Arnold Network for Satellite Image Classification(Cheon, 2024)	CNN + KAN	EuroSAT	Replaces MLP with KAN for efficient, interpretable learning	No graph or spatial relationship modeling
Geographical Area Classification on Satellite Images Using CNN(Wagh, 2024)	Modified CNN	Not specified	Simple CNN-based land cover classifier with dropout and pooling	No graph awareness or spatial context
Satellite Image Classification Using Deep Learning(Lakshmi, Prathik and Geetha, 2024)	CNN, ResNet	EuroSAT	ResNet improves spatial feature extraction in satellite data	Classification only. No localization or multi-level reasoning
CNN-Enhanced Multi-Scale Graph Attention Network (CEMSGAT)(Wang et al., 2024)	CNN + Multi-Scale GAT + SELF	Indian Pines, Salinas, WHU-HongHu	Fuses pixel-level CNN features with super-pixel GAT features using spatial transformations; excels in HSI	More complex architecture, focused on hyperspectral images.

Table 1: Existing Work

Table 1 illustrates that addressing satellite imagery classification has varying advantages and disadvantages associated with different methods of approach. CNN-only models are superb at finding local visual patterns (in the form of edges, textures, and shapes), but not at finding spatial relationships (between distant portions of the image). By contrast, GNN-only models are strong at representing non-Euclidean spatial relationships and global spatial context but have weaker performance at smaller-scale visual representations arising directly at image pixels. Such limitations are overcome through the concept of hybrid techniques, in which CNN-based feature extraction is complemented with graph-based

reasoning, with significant local and global cues being exploited. Though the methods are promising, most of the published hybrid works concentrate on categorising scenes or hyperspectral images, instead of being applied to the infrastructure detection task. Additionally, most previous research studies use a random split of data, which does not consider the spatial autocorrelation and tends to overestimate their model performances in realistic and geographically varied environments.

Although the domain of CNNs, GNNs, and hybrid deep learning techniques has advanced immensely, a fair number of gaps still exist in the literature. To start with, not much literature concentrates on infrastructure classification, i.e., the locality identification of residential, industrial, and highway regions, using RGB satellite imagery, although it is one of the most abundant information. Secondly, previous CNN-only architectures do not take into account spatial relationships between the neighbouring regions, which is known to lead to inconsistencies in the generation of predictions, especially on complex infrastructure layouts. Lastly, the various approaches previously found it difficult to use geographically held-out evaluation strategies, a critical need for the ability of model generalisation to unseen areas, and thus inflate the reported performances, making most unreliable in real deployment.

This research aims to address these shortcomings by formulating a hybrid CNN-GNN model, which targets the EuroSAT RGB data exclusively. High-level local features, e.g., shapes, textures, and edges, are extracted using the CNN backbone from the satellite images. These features are extracted and then used to build a graph representation with image patches representing nodes and spatial adjacencies represented as edges. The GNN module uses message passing operations to model global spatial relationships across the patches and thus fills in the local structure learning gaps of the CNN. The model is also tested using a geographically held out assessment scheme, where performance is put to the test on entirely novel geographic regions and giving a realistic performance metric on the generalisation to different geographic areas in infrastructure detection

4.10 Chapter summary

This literature review has discussed the trends in deep learning models to use in infrastructure classification in satellite images, which have advanced to use the relatively new GNN-based architectures, in addition to hybrid architectures that combine CNN and GNN. CNNs have become known to be very effective in the extraction of regional features and the recognition of patterns, though they have the drawback of not being able to model more intricate spatial relationships. GNNs provide better spatial dependency and contextual information-capturing capability, but can have trouble with extracting detailed visual features.

The development of a hybrid CNN-GNN architecture is a significant step forward towards resolving the weakness of any particular system. With the hybrid paradigms, these models combine the virtues of both paradigms and can perform more thorough visual learning of representation, simultaneously capturing local visual patterns and global spatial relations.

5. METHODOLOGY

This chapter presents the methodological approach followed to design, develop, and study a hybrid Convolutional Neural Network-Graph Neural Network (CNN-GNN) architecture for recognizing structures in satellite imagery. The data collection methodology is designed to include all procedural steps in a systematic, replicable, and academically rigorous process, such as the collection of datasets, the assessment of performance, and the evaluation of results.

The study objective is to categorise three infrastructure types, Residential, Industrial, and Highway, in the EuroSAT Sentinel-2 dataset using the complementary nature of CNNs and GNNs: CNNs to extract local features, and GNNs to model spatial relationships. The methodological process, therefore, involves a combination of principles in computer vision and graph-based deep learning, along with the preprocessing of data that leads to solutions of challenges of imbalance of data classes, pixel variability, and multivariate spatial patterns in remote sensing datasets.

This chapter presents research strategy and a project management approach, followed by the description of the dataset and data preprocessing steps, as well as the model architecture. The hybrid CNN GNN model is explained as per its constituents, flow of data, and the way it is integrated. The configuration of the training, choice of hyperparameters, and evaluation metrics are then described to make things clear and reproducible. There is also an ethical consideration, which pertains to the utilization of satellite information.

These steps are well-documented in this chapter, thereby providing a transparent framework upon which this implementation will occur to ensure the final Results and Analysis chapter is viewed in the right methodological manner.

5.1 Research Design and Proposed Approach

The proposed study has an experimental and practical research design in which it aims to examine the ability of a hybrid CNN-GNN architecture in the clusters of satellite pictures. It is a mixture of convolutional neural networks as a feature extractor and graph neural networks as a spatial relation modeller. The rationale behind the design lies in the earlier studies that indicated the inability of CNNs to excel at grasping the global context and the superiority of GNNs in describing the irregular structure of space.

The hybrid model was chosen instead of standalone CNN or GNN models because it is possible to get higher classification accuracies, or to be resistant to anomalous spatial patterns, or to have scale when utilizing standalone CNN or GNN in the real-world remote sensing location. The data science pipeline of the experiment involves choosing datasets, reading them, developing models, training, and testing models.

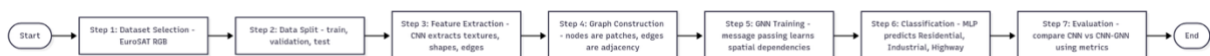


Figure 4: Proposed Approach

Step 1: Dataset Preparation

The Euro SAT RGB dataset is used for this study. Images are pre-processed and divided into training, validation, and test subsets following a geographically held-out split. This ensures the model is tested on unseen regions, improving its ability to generalize.

Step 2: CNN Feature Extraction

High-level spatial features are extracted by passing each satellite image through a pre-trained backbone (ResNet18) of CNN.

The CNN catches

- Textures of land surfaces,
- Geometry of buildings,
- Local residential, industrial, and highway patterns.

These are map features that are used as input attributes to the graph representation.

Step 3: Graph Construction

Using the CNN-generated feature maps, each image is converted into a graph:

- Nodes → represent super pixels or image patches.
- Edges → represent spatial adjacency (e.g., 4-neighbor connectivity) or visual similarity.
- Node features → come directly from the CNN embeddings.

This representation allows the model to capture both local texture information and global spatial relationships.

Step 4: GNN Training

The constructed graphs are fed into a GNN with two message-passing layers.

At each layer:

- Node features are updated by aggregating information from neighbouring nodes.
- Multi-hop reasoning enables the model to learn contextual dependencies between nearby and distant regions.

This stage enables the model to better understand the spatial layout of infrastructures

Step 5: Classification

The resulting graph-level representation of the GNN is fed into a Multi-Layer Perceptron (MLP) that is used to predict the infrastructure category:

- Residential
- Industrial
- Highway

A sigmoid activation converts logits to probabilities in binary classification.

Step 6: Model Evaluation

The trained hybrid model is evaluated using multiple performance metrics:

- Accuracy
- F1-score
- ROC-AUC
- PR-AUC

Spatial consistency between neighbouring regions

Performance is compared against a CNN-only baseline to demonstrate the effectiveness of the hybrid CNN-GNN architecture.

5.2 Project Management and Planning

The process of developing the hybrid CNN+GNN model required proper project management, as this would allow conducting it in an organized and on-schedule manner. The procedure would be an iteration-planned process where modifications were allowed to be introduced as intermediate experiment outcomes gave rise to the following process.

Scope Management

The project scope was well-developed at the beginning, which involved classifying residential, Industrial, and Highway infrastructure types based on the EuroSAT Sentinel-2 dataset. The overall work was broken into manageable steps: development of the literature review, obtaining and preprocessing the dataset, designing and training the CNN-GNN hybrid model, training and evaluating it, and lastly writing the dissertation report.

Time Management

A Gantt chart was created in order to plan milestones and monitor the progression of time. The phases, such as dataset preparation, model implementation, evaluation, and documentation, were subjected to certain timelines. Progress on the plan was systematically checked, and necessary corrections were made in case some tasks took longer time compared to the initially allocated time. This resulted in seamless project progress and timely completion, with no compromise on the quality of work.

Risk Management

Some of the risks envisaged during the project included the dataset imbalance, exposure to the limits of computation, and time lag in the convergence of models. They were mitigated in deploying such methods as data augmentation, oversampling to address the imbalance issue, tuning training parameters to accommodate the constraint of the available hardware, and other optimisation techniques to balance out the trade-off between training convergence and model stability. All this was designed into the project schedule through the building of buffer periods to cover unexpected hitches and to facilitate smooth project flow.

Quality Assurance

In order to achieve accuracy and reproducibility, all preprocessing, versions of the models, and hyperparameters used were well documented. The same random seeds were utilized when training the model, and various evaluation metrics (accuracy, F1-score, and ROC-AUC) were also used to obtain a complete picture of model performance.

Development Methodology

This was done using an iterative but Agile-inspired development process where there is flexibility to iterate on the preprocessing steps undertaken, hyperparameter tuning, and rearranging parts of the model based on experimental feedback.

Gantt Chart and Timeline

The project was planned to be implemented within a given period with milestones including the obtaining of the dataset, preprocessing of the dataset, model architecture design, training, assessment, and report.

5.3 Dataset Description

Data Source

The EuroSAT dataset used in the given research is a derivative of Sentinel-2 satellite images acquired by the European Space Agency. The data set contains images of 30 cities in different European countries (Helber et al., 2019). EuroSAT consists of multi-spectral images, which are distributed into ten classes of land use and land cover (LULC). In this study, three classes related to the infrastructure were selected. Namely, Residential (3000 Images), Industrial (2500 Images), and Highway (2500 Images).

Data Characteristics

Eurostat images are 64x64 pixels in size, and it has 13 spectral bands. The data are in two common forms:

1. **EuroSAT_RGB** – RGB bands in JPEG format.
2. **EuroSAT_MS** – 13 Sentinel-2 spectral bands in GeoTIFF format.

The dataset applied to this research was the EuroSAT_RGB, given that it is directly associated with the visual detection of infrastructure and is a good fit with standard computer vision procedures. It is also computationally easier to process than multispectral imagery, allowing the RGB version to be used in iterative deep learning experiments in the project timeline.

Rationale for Selection

The EuroSAT data set was selected based on several factors:

- **High-quality annotations** – Clear and well-defined class labels in the dataset do not necessitate the manual identification process of categories..

- **Open accessibility** – Freely available for use under open licensing.
- **Manageable dataset size** – With approximately 27,000 labelled and geo-referenced images, the dataset is large enough for deep learning tasks but small enough to allow efficient experimentation on available computational resources.
- **Relevance to research objectives** – Contains infrastructure-related categories that are very much aligned with the subject of the study.

The EuroSAT is simpler to interpret in terms of labeling than other datasets available, making it easier to prepare. This was a major consideration since the project had a time limitation. Moreover, the EuroSAT_RGB data format realizes the same complexity in pre-processing as multi-spectral products without losing the necessary spectral information in the classification direction of the infrastructure.

Data Collection

The EuroSAT dataset is based on Sentinel-2 satellite imagery and constructed to be classified as LULC (Helber et al., 2019). It has about 27,000 labelled images that have been fairly divided into ten classes. The three categories that were chosen in this study, namely Residential, Industrial, and Highway, would be taken out of the RGB version of the dataset.



Figure 5: satellite images from the Eurosat data set: Highway, Industrial, Residential

5.4 Data Preprocessing

Data Cleaning: Removal of any corrupted or duplicate images.

Image Resizing and Normalization: All images resized to a consistent format and pixel values normalized for model stability.

Data Augmentation: Techniques such as rotation, flipping, brightness adjustment, and contrast enhancement to address class imbalance.

Class Balancing: Oversampling underrepresented classes or applying class weights during training.

Train-Validation-Test Split: 70-15-15 split ratio, ensuring balanced representation of all classes across splits.

5.5 Model Design

CNN Component

Local feature extraction of input images was done with a convolutional neural network. The CNN has several convolutional and pooling layers, and the resulting feature vectors can be flattened to form a feature vector that can be used to construct a graph.

Graph Construction

The superpixels or image patches were considered as nodes, the edges corresponding to neighbourhood in either the spatial or feature space. The CNN feature vectors act as the attributes of the node, a part of the graph.

GNN Component

The relationships between nodes were modeled with a basic message passing Graph Neural Network. Features of the nodes were updated at every iteration, adding neighbourhood node features and then performing a learnable transformation. Several iterations had the effect of the network grasping more and more complex spatial relations.

Hybrid Integration

Sequential integration was applied, and input CNN feature maps were fed into the GNN to reason about relations, followed by classification. The structure used is in such a way that visual & spatial context both play their part in predictions.

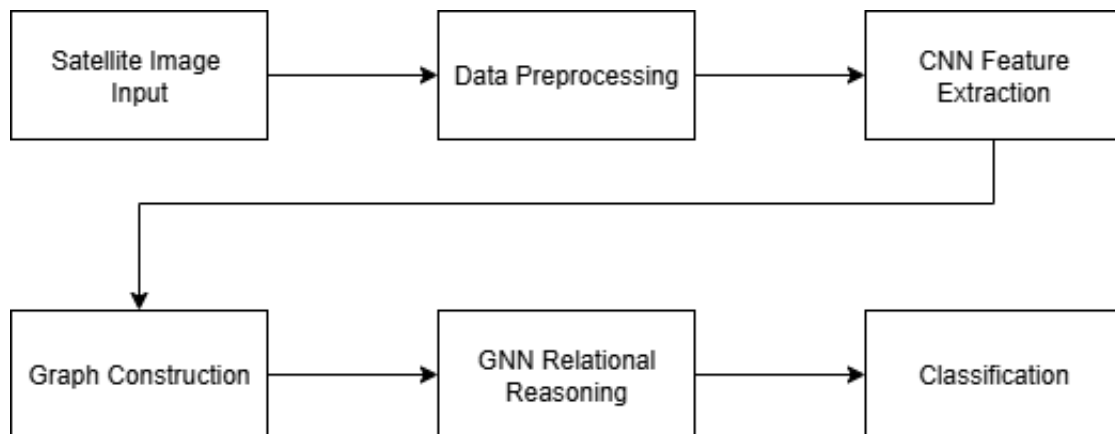


Figure 6: high-level architecture diagram

5.6 Exploratory Data Analysis

Class Distribution

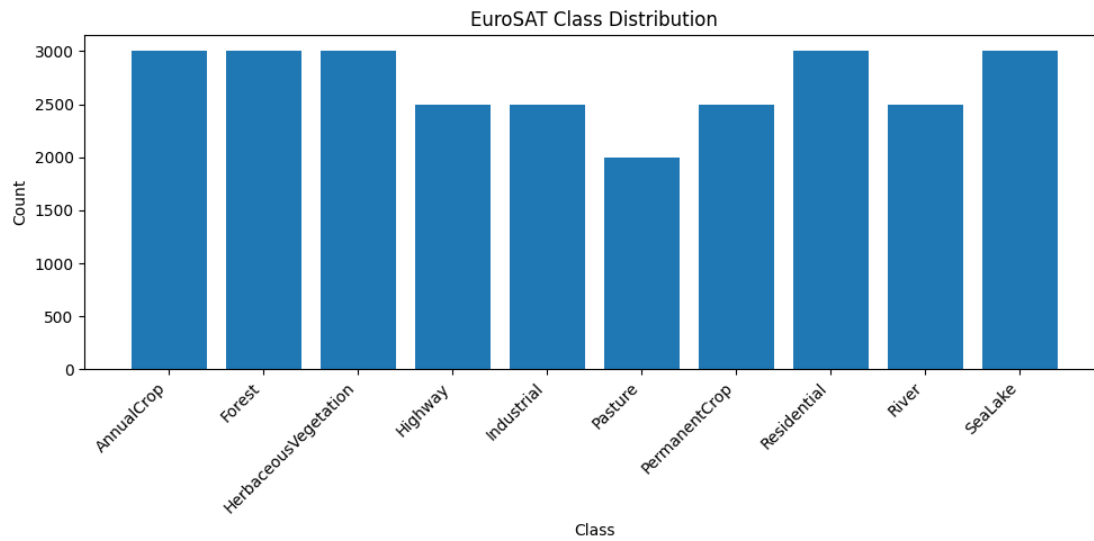


Figure 7: EuroSAT Class Distribution (Code Snippets are in appendix)

As shown in Figure 7: EuroSAT Class Distribution (Code Snippets are in appendix) the percent distribution of the 10 classes used by the original EuroSAT data. The data is balanced to a certain extent, where the larger classes have 2,500-3,000 samples. Annual Crop, Forest, Herbaceous Vegetation, Residential, and Sea Lake consist of 3,000 pictures each, compared to Highway, Industrial, Permanent Crop, and River, which have 2,500 pictures each, respectively. The class with the fewest number of samples, Pasture, comprises 2,000 samples. Such a balance is used to reduce the bias of classes in training in the original multi-class scenario. Nonetheless, in the binary classification problem utilized in this study, particular classes, such as Highway, Industrial, and Residential, are placed under Infrastructure, while the rest of the classes are categorized as Non-Infrastructure. The result is an imbalanced dataset that must be taken into consideration when designing and evaluating any model.

Random Samples

Figure 8 shows a randomly picked subset of images in the EuroSAT dataset with the original classes span. It is evident in the samples that land cover and land use types are very diverse, featuring such elements as urban areas, transportation networks, agricultural fields, forests, water bodies, and barren land. This diversity is indicative of the data set's capability to match a wide range of spectral and spatial characteristics that can be encountered in satellite imagery. In visual inspection, one can note that the sharing of colour and texture properties of some classes would be highlighted- e.g., some agricultural fields might have a similar characteristic with barren land, and built-up areas might have good areas of vegetation cover. These apparent similarities in different categories show that the classification may encounter certain problems and provide an incentive to apply means of feature extraction able to reflect not only the spectral signature but also the spatial configuration.

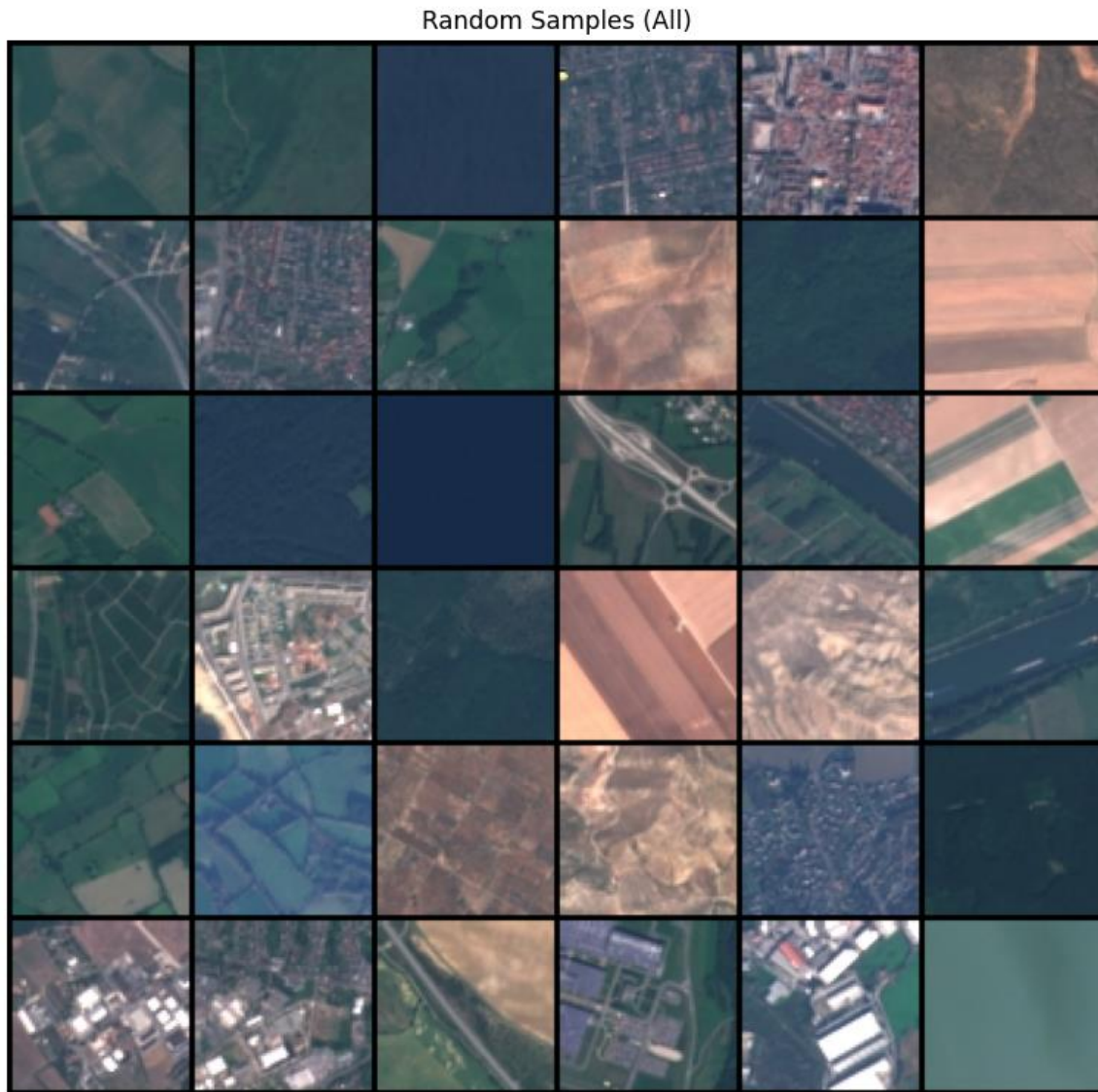


Figure 8: Random Sample Images From EuroSAT Dataset

Figure 9 shows 12 randomly chosen examples of each class in the binary dataset, where Non-Infrastructure images are shown on the upper row and Infrastructure on the bottom. There is a significant difference in the content of Non-Infrastructure and Infrastructure samples, with the first area mostly representing agriculture fields, forests, and other natural landscapes, and the latter area representing highways, industrial areas, and residential areas. The variance in classes notes the range of spatial patterns, texture, and spectrum signatures to be learned by the model to differentiate. Existence of visually related patterns belonging to the different classes: e.g., road-like or vegetation-like structures in agricultural regions or patches in industrial regions, highlights the sophistication of the classification task and the necessity of integration of both visibility and spatial environments in modelling.



Figure 9: 12 Samples per Binary Class (Top: Non-Infrastructure, Bottom: Infrastructure)

Pixel Intensity Histograms (R, G, B)

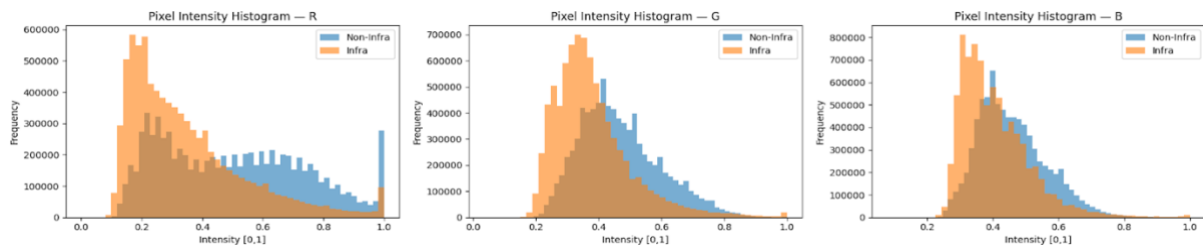


Figure 10: Pixel Intensity Histograms (R, G, B) (Code Snippets are in appendix)

Figure 10 shows the histograms of pixel intensity of Red (R), Green (G), and Blue (B) channels of the binary dataset, the comparison between Infrastructure and Non-Infrastructure classes. Each of the intensities is normalized to $[0,1]$.

In the Red channel, Infrastructure images exhibit a significant maximum at low intensities (0.15 - 0.3) corresponding to a darker or dull red color that is characteristic of concrete, asphalt, and rooftop industrial buildings. On the contrary, Non-Infrastructure images show a wider dispersal reaching even more into red intensities, which is dependent on soil and vegetation, perhaps.

In the Green channel, Non-Infrastructure images dominate in the medium intensity (0.4-0.6) due to the prevalence of vegetation in rural or natural scenery. There is an increased concentration in the infrastructure images at low green intensities.

In the Blue channel, the frequencies of the two classes are highest in mid-level intensities (0.35-0.45), though Non-Infrastructure also has the highest frequency, probably because it includes water bodies and sky.

Channel	Infrastructure Peak Intensity Range	Non-Infrastructure Peak Intensity Range	Interpretation
Red (R)	0.15 - 0.30	0.20 - 0.60	concrete, asphalt, rooftops dominated by darker/muted reds, Non-Infrastructure broader because of soil and vegetation tones.

Green (G)	0.25 - 0.40	0.40 - 0.60	Non-Infrastructure is greener, richer in vegetation colours; Infrastructure is less green with reduced green intensities.
Blue (B)	0.30 - 0.45	0.35 - 0.50	Both peak in mid-blues; non-infrastructure slightly higher due to water bodies and sky.

The differences in pixel intensity distributions between channels that we find indicate that spectral data alone contains informative characteristics that can be used in order to separate Infrastructure and Non-Infrastructure. This is evident using the green channel intensities in non-infrastructure images, which have elevated values in the green region as compared to the red, hence representing vegetative growth, and Infrastructure images are more darkened in the red region or have low green intensities signifying the built or urban region. Nevertheless, the overlapping distributions of each of the three channels point to that pixel-level colour features might not be sufficient to fully separate the classes. The overlap justifies the necessity of a CNN-GNN hybrid model so that the CNN model extracts the spectral-textural features and the GNN model represents the spatial relationships, and the network could learn to handle the scenario where the two categories have a visual similarity.

PCA and t-SNE

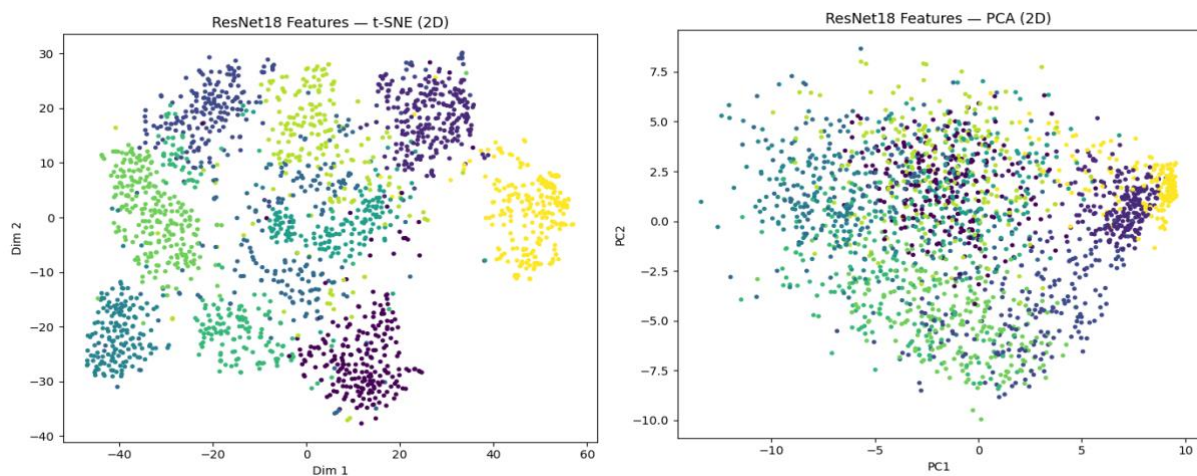


Figure 11: PCA and t-SNE projections for the original 10-class EuroSAT dataset (Code Snippets are in appendix)

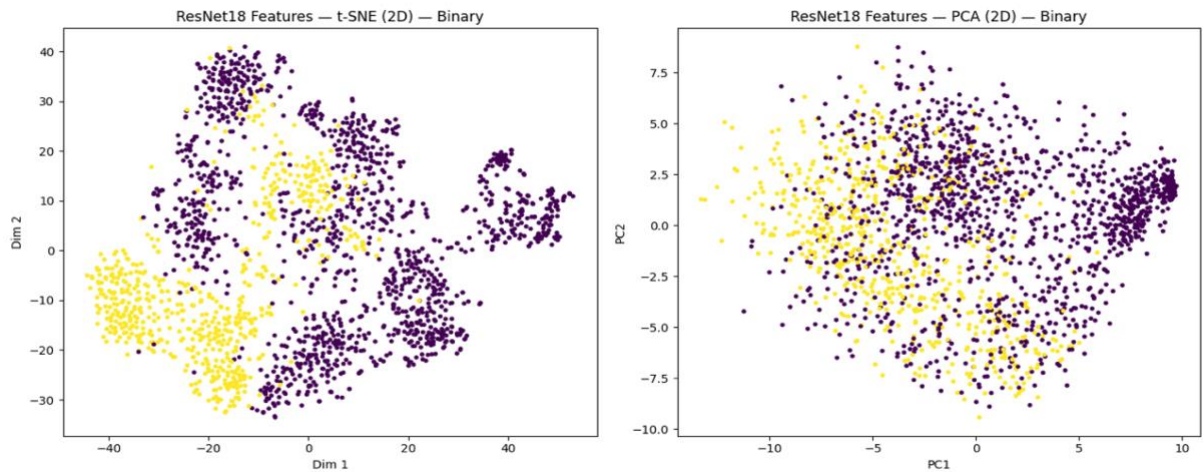


Figure 12: PCA and t-SNE projections for the Infrastructure, non-Infrastructure class EuroSAT dataset (Code Snippets are in appendix)

Figure 11 illustrates PCA and t-SNE projections on the original EuroSAT dataset with the 10-class labeling. Already, it is apparent that there is a large portion of overlap between many of the classes in PCA and that there are very few resulting distinct clusters (e.g., SeaLake). On the contrary, the work of t-SNE creates denser and more separable clusters; however, some agricultural and urban classes still overlap.

The same is presented in Figure 12, but the analysis is on a binary classification case (infrastructure vs non-infrastructure). The PCA projection displays the same way in that much mixing occurs among the two classes to display the same global characteristic visual features. The t-SNE projection proves that there are local clusters of both classes, but the overlap is evident, especially in situations where visual patterns are close to each other (e.g., highways in the countryside or industrial regions that look like crop fields).

This discussion emphasizes the fact that the original and the binary classification problem have an inherent visual similarity among the classes, and thus, it is hard to separate the classes. These facts foster the selection of a hybrid CNN + GNN model, which may use local similarities of features as well as spatial relations to enhance class differentiation.

5.7 Evaluation Metrics

It is important to determine the performance of the suggested hybrid CNN-GNN model using a series of quantitative assessment measures. The choice of these metrics aimed at having an in-depth insight into the performance of the model at both overall and per-class levels in order to be able to compare it directly with a baseline CNN-only model.

Accuracy is a measure of how well the classes in the data are classified from the entire amount of data. Although it provides a fast understanding of performance, it is inaccurate when dealing with class imbalance because high accuracy could be obtained just because the majority of the classes are being predicted.

$$\text{Accuracy} = \frac{T_P + T_N}{T_P + T_N + F_P + F_N}$$

Where:

TP – True Positives (correctly predicted positive samples)

TN – True Negatives (correctly predicted negative samples)

FP – False Positives (incorrectly predicted positive samples)

FN – False Negatives (incorrectly predicted negative samples)

Precision measures how many positive examples were identified correctly, and as a total number of examples that were identified as positive. It plays an especially significant role in the presence of high costs of false positives.

$$\text{Precision} = \frac{T_P}{T_P + F_P}$$

Recall (also Sensitivity or True Positive Rate) is the ratio between the number of positive predictions among all actual positives. It becomes imperative when one wants to avoid a false negative (where a positive case is missed) in a case where the cost of a false negative is high.

$$\text{Recall} = \frac{T_P}{T_P + F_N}$$

The F1-score gives a balanced measure whenever false positives and false negatives are of interest in estimation, since it is the harmonic mean of Precision and Recall.

$$\text{F1 measure} = \frac{2 \times Pre \times Rec}{Pre + Rec}$$

The metric particularly helps where there is an imbalance in the dataset, in which case, raw Accuracy would not be as accurate.

Area under the Receiver Operating Characteristic (AUC-ROC): This will assess whether the model can make predictions above-average based on the different threshold settings it produces between classes. ROC curve is a graph of the True Positive Rate (Recall) against the False Positive Rate.

AUC can be between 0.5 (no discrimination, or equivalent to chance guessing) to 1.0 (perfect discrimination). As AUC increases, the performance of classification is higher at various thresholds.

The benchmark model was comparison-wise calculated with the benchmark CNN-only model. To ensure a fair comparison, we trained both models with an identical split

(train/validation/test), identical preprocessing pipeline, and identical optimization hyperparameters.

The confusion matrix is in the form of a table indicating how many correct predictions were made and incorrectly made per class. It assists in finding certain misclassification patterns among the categories. In the case of a multi-class classification problem, the confusion matrix is an $n \times n$ matrix with n being the number of classes.

5.8 Implementation Tools and Libraries

This project was implemented by leveraging the power of assembled development, programming, and productivity tools and environments to support efficient model development, experiment tracking, and documentation.

Programming Language and Environments

- **Python** – Used to implement the architecture hybrid CNN-GNN as it is one of the most supported programming languages to be used in such a system due to its high capabilities in deep learning, scientific computing, and data processing.
- **Jupyter Notebook** – Interactive experimenting, step debugging, and interactive visualisation of intermediate model state over time are taught and practiced using it.
- **Visual Studio Code** – Served as the main integrated development environment (IDE) for writing and organising Python scripts, as well as managing the overall codebase.
- **Visual Studio** – Used for additional coding tasks and environment management where required.

Deep Learning and Data Processing Libraries

- **PyTorch** - A generalized and popular deep learning framework utilized to institute, teach, and validate the CNN part of the hybrid outline. It had the benefit of its dynamic computation graph, which was beneficial to iteratively experimenting and making architectural changes.
- **PyTorch Geometric** - A highly specialized extension library to PyTorch, allowing to work with graph-structured data, which was used to implement and train the GNN part of the hybrid architecture. It offered well-tuned data loaders, layers, and message-passing that are critical in graph-based learning.
- **NumPy** - A core Python library of numerical usage that is used in array manipulation, mathematics applications, and in the effective management of intermediate data structures.
- **Pandas** - A high-performance data manipulation library, used for managing metadata, dataset statistics, and experimental logs.
- **Matplotlib** - A versatile visualization library, used for plotting training curves, evaluation metrics, and dataset class distributions.
- **Seaborn** - A library based on Matplotlib, which is designed to create plots and heatmaps, especially useful in creating confusion matrices, and in the creation of correlation plots.

Data Storage and File Management

- **Google Drive** - was used as the primary cloud storage for data sets, learned models, and running results, and for the documentation
- **OneDrive** - Used for additional cloud backup and file sharing across devices.

Documentation and Collaboration Tools

- **Microsoft Word** - Used for drafting and formatting the dissertation document.
- **Google Docs** - real-time editing and makes written documents accessible across many devices.

Such a composition of tools offered a flexible and powerful working process that was efficient in terms of development, safe storage, and simplified documentation during the entire project cycle.

5.9 Ethical Considerations

Responsible use of data and academic integrity and fairness in designing the model were observed in this research, as provisions on ethical considerations were taken throughout the development and implementation process.

Data Privacy and Licensing

The dataset utilized in the present study is publicly available and released using an open-access license, EuroSAT. No personally identifiable data is included in the data, and there are no issues regarding privacy. The data was applied with the purpose of academic research.

Responsible Use of Satellite Imagery

Satellite imagery can be utilized in many areas and can be used in surveillance and commercial exploitation; however, this research is specifically based on infrastructure classification in an academic sense only. The approach and findings are neither applied to nor built to be used in any practice that will seek to violate the privacy of any individual, including those that would enable unethical surveillance.

Bias and Fairness

Caution was taken to minimize the effect of bias in the predictions made by the model by reducing the class imbalance among the data in the dataset with data augmentation and balancing the evaluation metric. The impact of model predictions might even affect the decision-making process during urban planning or monitoring the environment, so one crucial concern was to reduce the bias and ensure unbiased and fair predictions.

Reproducibility and Transparency

The whole methodological part involved in dataset preparation, preprocessing, the definition of the model, and its evaluation is described in references to enable this exercise to be reproduced. Data, code, and experimental outputs are stored safely.

Following these principles, the study was performed for academic integrity, considering the considerations of ethical implications of the use of satellite imagery, fairness, and transparency in artificial intelligence uses.

5.10 Chapter Summary

This chapter has outlined the methodological approach in developing, applying, and testing the proposed hybrid CNN GNN model for classifying the infrastructure in satellite images. The design of the research has an experimental and applied character and combines the methods of convolutional neural networks that were used to extract the local features with graph neural networks that were used to model the spatial relations.

A large project such as it was handled through the iterative development process and included a series of clearly laid-out milestones as well as a variety of programming tools, cloud storage solutions, and documentation programs. The high-quality annotations and relevance with the classification of Residential, Industrial, and Highway infrastructure to the classification task led to the choice of the EuroSAT dataset. The preprocessing data was cleaned, resized, normalised, augmented, and balanced since it was required to enhance the generalisation of the model and due to dataset imbalances.

The sequential integration structure is a form of hybrid CNN-GNN model that utilizes a feature extraction mechanism through a CNN backbone substructure, constructs the graph by forming nodes and their connections, and employs a GNN substructure to process relational reasoning. Model training and optimisation approaches were outlined, and the evaluation measures were chosen that would be used to indicate the overall and per-class performance. Ethics were also covered to address responsibility, fairness, and reproducibility of the data use.

Results and Analysis. The following chapter will state the results of the experiments on the baseline CNN-only model as well as the proposed CNN and GNN hybrid architecture, and compare the performances of the two using the set evaluation metrics. The results will be explained to the research question, whereby the strengths, limitations, and implications of the study method will be described.

6. RESULTS

7. CONCLUSION

8. FURTHER WORK

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10. Appendix

Appendix 1

Link to the source code:

Appendix 2

Python code snippets for the exploratory data analysis

```
# Pixel Intensity Distribution (per channel)

subset = random.sample(range(len(ds)), k=min(5000, len(ds)))
vals = []
for i in subset:
    x, _ = ds[i]
    vals.append(x.view(3, -1).numpy())
vals = np.concatenate(vals, axis=1) # [3, N]

for ch, name in enumerate(["R", "G", "B"]):
    plt.figure(figsize=(6,4))
    plt.hist(vals[ch], bins=50)
    plt.title(f"Pixel Intensity Histogram - {name}")
    plt.xlabel("Intensity [0,1]"); plt.ylabel("Frequency")
    plt.tight_layout(); plt.savefig(os.path.join(OUT_DIR, f"hist_{name}.png"), dpi=200); plt.show()

# Image Characteristics (resolution, overall channel stats, pixel range)

sample_idx = random.sample(range(len(ds)), k=min(2000, len(ds)))
heights, widths, px_min, px_max = [], [], [], []
for i in sample_idx:
    x, _ = ds[i]
    h, w = x.shape[1], x.shape[2]
    heights.append(h); widths.append(w)
    px_min.append(float(x.min())); px_max.append(float(x.max()))

pd.Series([f"{w}x{h}" for w,h in zip(widths, heights)]).value_counts().sort_index() \
    .to_csv(os.path.join(OUT_DIR, "image_sizes_counts.csv"))

plt.figure(); plt.hist(heights, bins=10); plt.title("Height (px)"); plt.xlabel("px"); plt.ylabel("freq")
plt.tight_layout(); plt.savefig(os.path.join(OUT_DIR, "height_hist.png"), dpi=200); plt.show()

plt.figure(); plt.hist(widths, bins=10); plt.title("Width (px)"); plt.xlabel("px"); plt.ylabel("freq")
plt.tight_layout(); plt.savefig(os.path.join(OUT_DIR, "width_hist.png"), dpi=200); plt.show()

# Pixel Intensity Histograms per binary class

def pixel_histograms_by_binary(dataset, per_class_samples=2500):
    idxs_0, idxs_1 = [], []
    # stratified sampling
    for i in range(len(dataset)):
        _, yb = dataset[i]
        if yb == 0 and len(idxs_0) < per_class_samples:
            idxs_0.append(i)
        elif yb == 1 and len(idxs_1) < per_class_samples:
            idxs_1.append(i)
        if len(idxs_0) == per_class_samples and len(idxs_1) == per_class_samples:
            break

# Feature Space Analysis (ResNet18 embeddings) - PCA & t-SNE (binary colors)

DEVICE = "cuda" if torch.cuda.is_available() else "cpu"
resnet = models.resnet18(weights=models.ResNet18_Weights.IMAGENET1K_V1)
resnet.fc = torch.nn.Identity()
resnet = resnet.to(DEVICE).eval()

feat_tfms = transforms.Compose([
    transforms.Resize((224,224)),
    transforms.ToTensor(),
    transforms.Normalize(mean=[0.485,0.456,0.406], std=[0.229,0.224,0.225]),
])
```

```

# PCA (2D)
pca = PCA(n_components=2, random_state=SEED)
p2 = pca.fit_transform(feats)
plt.figure(figsize=(7,6))
plt.scatter(p2[:,0], p2[:,1], s=8, c=labs_bin)
plt.title("ResNet18 Features - PCA (2D) - Binary")
plt.xlabel("PC1"); plt.ylabel("PC2")
plt.tight_layout(); plt.savefig(os.path.join(OUT_DIR, "pca_2d_binary.png"), dpi=200); plt.show()

# t-SNE (2D)
ts = TSNE(n_components=2, init="pca", learning_rate="auto", perplexity=30, random_state=SEED)
t2 = ts.fit_transform(feats)
plt.figure(figsize=(7,6))
plt.scatter(t2[:,0], t2[:,1], s=8, c=labs_bin)
plt.title("ResNet18 Features - t-SNE (2D) - Binary")
plt.xlabel("Dim 1"); plt.ylabel("Dim 2")
plt.tight_layout(); plt.savefig(os.path.join(OUT_DIR, "tsne_2d_binary.png"), dpi=200); plt.show()

```